Vibe Shuffle: Baseline-Relative Multimodal Sensing for Adaptive Next-Song Selection

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Abstract

Music recommender systems learn durable preferences, but the track that fits can change from one moment to the next. *Vibe Shuffle* examines whether lightweight camera and cardiac observations can inform next-song selection without being presented as a readout of emotion. It expresses each channel as change from a personal reference, collects successive relative Valence–Arousal points during playback, and uses their mean to select the nearest eligible track from the matching region of a fixed catalog. An exploratory within-participant crossover compared this policy with Random selection from the same catalog. Mood fit was higher on average under Vibe Shuffle, while liking changed little; both estimates were uncertain and did not establish a recommendation benefit. The project provides an inspectable research system that connects personal references, multimodal observations, and song choice. It also defines the validation needed before evaluation as a session-aware re-ranking layer in a music service.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Information systems** → *Recommender systems.*

Keywords

affective computing, music recommendation, relative affect, valence and arousal, heart-rate variability, facial cues, baseline-relative adaptation

1 Motivation

1.1 Contextual music choice

Music choice is contextual. A track that suits exercise may be unwelcome during a quiet break. People use music to regulate mood and activation [11, 25]. Most recommender systems prioritize durable signals such as listening history, favorite artists, similar users, and item features. Those signals characterize usual preference and capture momentary fit only indirectly, a known challenge in contextual music recommendation [12, 26]. The resulting gap is practical: the listener may like a song in general without wanting it next.

Explicit mood selection can address that gap, but it also interrupts an otherwise continuous experience. Automatic adaptation offers another route. Its observations, however, are uncertain. Facial behavior depends on pose, lighting, culture, context, and deliberate expression [2, 5, 14]. Cardiac timing varies with respiration, posture, movement, fitness, medication, and sensor quality [15, 23, 27]. Neither channel directly identifies emotion.

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1.2 Baseline-relative next-song adaptation

Vibe Shuffle studies whether such observations can guide the next song without being presented as absolute emotion labels. During setup, the listener provides short personal camera and cardiac references. Later observations are expressed as changes from those references. The horizontal axis represents relative Valence evidence; the vertical axis represents relative Arousal evidence. Each point falls into a Tense, Energetic, Melancholic, or Calm recommendation region. These names identify catalog regions, not the listener's emotional state.

The catalog uses song Valence and Energy coordinates. Vibe Shuffle averages the recent trajectory, retains unplayed tracks in its region, and selects the nearest eligible track. Random draws from the same remaining catalog without the trajectory. The adaptive chain is explicit: personal references, relative observations, trajectory, region, and nearest available song.

1.3 Research question and contributions

Research question. How large and uncertain are the differences in participant-rated mood fit between the baseline-relative policy and Random selection from the same fixed catalog? Liking is treated as a secondary outcome.

Research contributions.

- **Baseline-relative operationalization.** Personal reference periods center camera and cardiac observations before they enter the relative Valence and Arousal coordinates.
- **Specified selection policy.** Event-driven observations form a trajectory; its mean defines a region, and distance selects the nearest unplayed song from that region.
- **Paired evidence.** The exploratory crossover estimates mood-fit and liking differences against Random selection from the same frozen catalog under matched exposure and rating procedures.

Mood-fit estimates favored Vibe Shuffle, whereas liking barely changed; both intervals crossed zero. A service-level evaluation would require validated sensor mappings, complete trial records, preregistration, and a dynamic catalog.

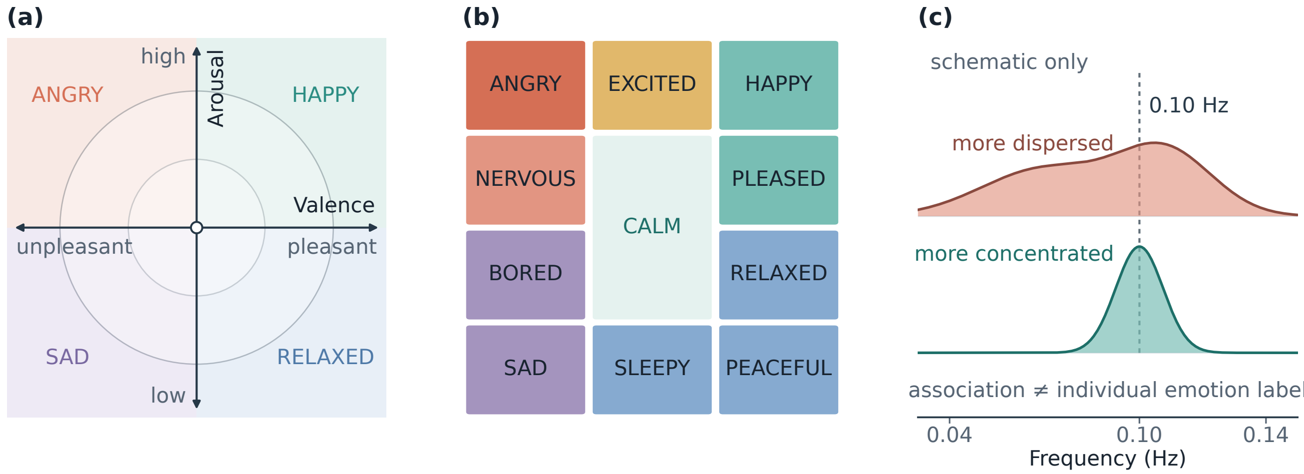

Figure 1: Conceptual foundations. Panels (a)–(b) are original redrawings adapted from Helmholz et al. [9], who operationalized Russell's circumplex [24] and Thayer's mood–arousal model [28]. Panel (c) is an original schematic adapted from the group-level frequency distinction reported by Balaji et al. [4]; it contains no empirical observations.

2 Background

2.1 Valence–Arousal as a retrieval space

Dimensional affect models provide a shared vocabulary for listeners and music. Russell's circumplex represents Valence, from unpleasant to pleasant, and Arousal, from low to high activation [22, 24]. Thayer's account adds complementary categories for energetic and tense activation [28]. Both dimensional and categorical models occur in music-emotion research, and results depend on the selected model, labels, and stimuli [7]. Vibe Shuffle adopts this vocabulary as a coordinate for recommendation, not as a complete account of experience.

Figure 1 contrasts continuous coordinates, a categorical vocabulary, and HRV-frequency context. The policy outputs neither emotion words nor coherence profiles. Its coordinates encode within-listener change, which does not make facial or cardiac observations emotion-specific.

Songs occupy a parallel Valence–Energy space. Song Valence describes conveyed positivity, whereas Energy describes intensity and activity. Prior interfaces have used these catalog features as counterparts of affective Valence and Arousal [9]. Vibe Shuffle follows that convention while keeping the constructs distinct: musical Energy is not physiological Arousal. Music-emotion coordinates depend on subjective annotation and compress a course that changes over time [1, 30].

Splitting both axes at 0.5 yields four recommendation regions. The listener's relative position selects a region, region membership determines eligibility, and Euclidean distance ranks the eligible songs. The coordinate is thus a controlled retrieval space rather than a ground-truth emotion measure.

2.2 HRV as relative evidence

HRV describes beat-to-beat variation, and RMSSD summarizes short-term RR-interval variation [27]. Recording length, respiration, movement, posture, fitness, medication, contact quality, and vendor processing limit interpretation [15, 23]. Vibe Shuffle uses heart rate and RMSSD only as baseline-relative Arousal evidence. Its custom spectral-peakedness field is diagnostic, differs from the group-level coherence measure in Balaji et al. [4], and never affects selection.

3 Related Work

Music can evoke, express, maintain, or regulate affect [11]. Emotion-aware systems use explicit or inferred context to constrain the next track. *Feel the Moosic* maps a listener-selected affect position to Spotify Valence–Energy ranges [9]. Vibe Shuffle instead derives the coordinate from baseline-relative camera and cardiac observations over a listening window.

Janssen et al. learned listener-specific relations between music and physiological response and used them to steer an affective music player toward a goal [10]. Ayata et al. estimated Valence and Arousal from wearable galvanic-skin-response and photoplethysmography measurements before recommending music [3]. Tran et al. matched a facial estimate with nearby tracks in Spotify feature space [29]. Together, these systems map personal or inferred context to music, but do not show that selected songs improve momentary fit.

The path from observation to selection can hide consequential assumptions. Explanations may support transparency, trust, effectiveness, or decision support [21]; perceived quality also depends on effort, privacy, and context [13]. A reviewable policy should expose what was observed, how observations were summarized, and why one song became eligible.

Among the systems reviewed here, few expose a baseline-relative path from multimodal observations to a reviewable next-song decision while holding the catalog, exposure, and rating procedure constant. Vibe Shuffle addresses that gap with a personal reference, an event-driven listening trajectory, a fixed region-and-distance rule, and a Random comparator drawn from the same frozen catalog.

Its distinctive claim concerns the specified and testable decision path; the exploratory ratings do not establish superiority.

4 Concept Overview

Three objectives guided the concept. First, interpretation should be personal: observations are expressed as changes from the listener's own reference rather than population thresholds. Second, the decision should remain inspectable: the research record should expose the path from observations to the selected song. Third, adaptation should be conservative: sensor outputs guide recommendation but are not shown as the listener's emotion.

The project tests matching rather than mood regulation. It does not assume that an energetic track should calm activation or that positive Valence should repair a negative state. Maintaining, intensifying, and shifting would require separate policies.

The archive preserves the final research system but not sketches, interviews, or a dated prototype history. Its visible refinements are a personal reference for within-listener change, a trajectory rather than one sample, a fixed region-and-distance rule, and a Random comparator from the same catalog. A later audit separated diagnostic coherence from selection, documented provenance and limitations, and clarified the claim boundary. These are specification and reporting changes; no qualitative feedback log supports participant-driven redesign. Optional inputs and explicit signal-path export let a session continue when a channel is missing.

Vibe Shuffle follows a six-step loop: acquire personal references, collect camera and cardiac observations during playback, express them as relative Valence–Arousal points, summarize the trajectory, choose the nearest eligible song in the corresponding catalog region, and record the rating and decision trace. Adaptation occurs between tracks, so brief fluctuations cannot interrupt the current song. Figure 2 shows this loop.

The intended use is scientific evaluation of baseline-relative adaptation. A future music service could test the policy as a re-ranking layer above a conventional preference model. The present project neither implements nor validates that deployment.

5 Signal & Pipeline

5.1 Personal references

The multimodal input has two distinct channels. A camera supplies visual and behavioral cues; an optional wearable supplies physiological timing. The camera reference runs for up to 14 seconds and may be skipped. The cardiac reference records 120 seconds of device-produced heart rate and RR intervals. Subsequent observations are centered on these personal values. The references do not define a neutral or true emotional state; they provide local comparison points for the session.

The camera supplies visible evidence primarily for relative Valence; cardiac change supplies relative activation. Both are accessible through commodity interfaces, but each mapping requires separate validation. Figure 2 shows how the relative inputs enter the trajectory and catalog decision.

5.2 Relative Valence and Arousal signals

MediaPipe Face Landmarker runs locally and returns facial blendshape activations [17]. A fixed heuristic combines selected smile, cheek, brow, eye, jaw, frown, and mouth cues into positive, negative, and tension-related scores. Samples scheduled at 120 ms intervals are debiased against the reference and smoothed across frames. More positive visible evidence moves relative Valence right; negative or tension-related evidence moves it left. A slowly adapting reference follows longer changes. If no face is detected, Valence returns to the center.

Visible movement and nodding can also add positive-Valence and upward-Arousal evidence. The channel represents change in observable behavior under a fixed rule, not felt emotion. Lighting, pose, occlusion, morphology, culture, deliberate expression, dancing, and repositioning may change the output [2, 5, 14]. The complete coefficients and gates are retained in `src/expressionModel.js`.

A compatible Bluetooth Heart Rate Service device supplies beats per minute and, when available, RR intervals. The research system has no raw electrocardiogram and cannot inspect the device's beat detector. Implausible or abruptly changing intervals are rejected before variability is calculated. The reference stores median heart rate, the root mean square of successive differences (RMSSD), and robust spread estimates.

During listening, higher-than-reference heart rate and lower-than-reference RMSSD provide upward Arousal evidence; changes in the opposite direction provide downward evidence. Heart rate receives the larger influence, and RMSSD is damped when the indicators disagree. At least 20 accepted RR intervals are required for the full estimate. RMSSD is a recognized short-term variability summary, but duration, respiration, posture, movement, fitness, medication, contact quality, and vendor processing affect its interpretation [15, 23, 27]. Agreement between 120-second and longer RMSSD measurements has been reported under controlled resting conditions, not short listening windows with movement [19].

The export also records SDNN and a custom spectral-peakedness field called physiology coherence. It differs from the coherence measure associated with post-session emotion labels in large-scale biofeedback data [4]. It is diagnostic only and does not affect song selection.

5.3 Fusion, quality, and data flow

The pipeline is acquisition, plausibility filtering, personal-reference centering, smoothing, quality assessment, and coordinate fusion. Usable cardiac evidence sets the Arousal base, while camera movement can add upward evidence. Camera movement supplies an upward-only fallback when cardiac data are unavailable. If neither channel is usable, the coordinate returns to the central reference point.

Raw camera frames, facial landmarks, and Bluetooth notifications remain in browser memory. A locally downloaded CSV contains derived summaries, reference status, signal source, quality flags, timing, selected-song metadata, and ratings. It excludes frames, landmarks, raw packets, waveforms, complete trajectories, personal facial-reference values, and the session seed. This is a data-minimizing local export, not a formally privacy-preserving system. Spotify playback, MediaPipe assets, and hosting still require network traffic.

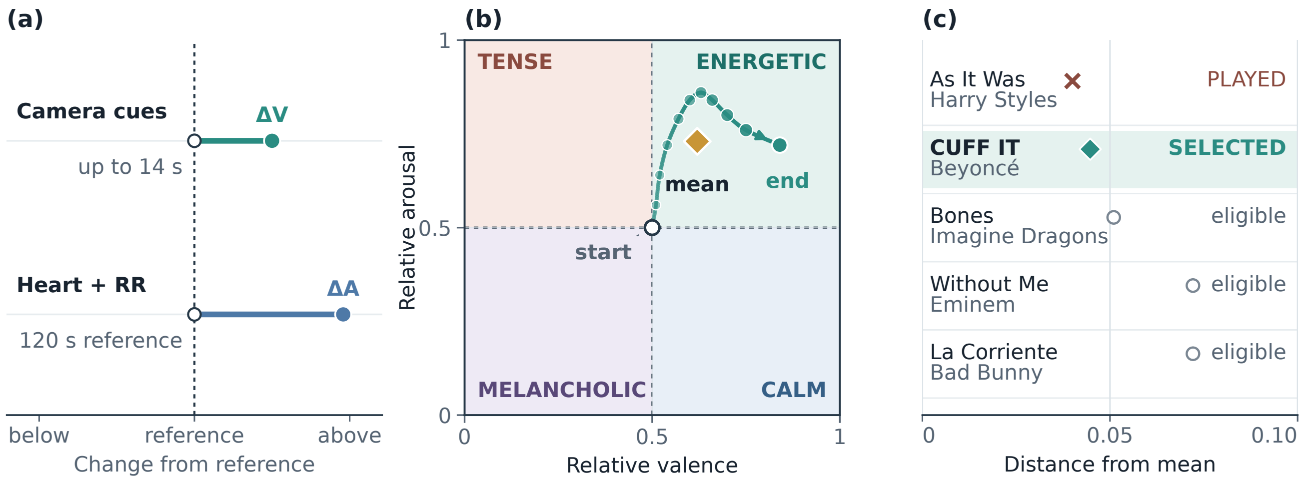

Figure 2: Baseline-relative sensing, trajectory aggregation, and catalog selection. (a) Signed camera and cardiac changes use personal references. (b) The arithmetic mean (gold diamond) summarizes the trajectory between its open start and filled endpoint. (c) *As It Was* is nearest but hypothetically marked as played, so *CUFF IT* is the nearest eligible choice. Panels (a)–(b) are schematic; panel (c) uses catalog-derived candidates and distances with an illustrative played state. Coherence is logged but not used for selection.

6 Inference & Adaptation

6.1 Trajectory and catalog target

During confirmed playback, a change in the fused coordinate appends a point to the trial trajectory

$$\mathcal{T} = \{(V_1, A_1), \ldots, (V_m, A_m)\}.$$

At rating time, the arithmetic mean

$$(\bar{V}, \bar{A}) = \frac{1}{m}\sum_{i=1}^{m}(V_i, A_i)$$

becomes the target for the next Vibe choice. Aggregation reduces dependence on one observation but is event-weighted, because points are added when the coordinate changes rather than at a fixed rate. The export records sample count and listening duration for review.

Splitting both axes at 0.5 produces four catalog regions: Energetic, Calm, Tense, and Melancholic. These names identify Valence–Energy partitions; they do not classify the participant. Listener Arousal and song Energy are operational counterparts for retrieval, not equivalent constructs [9, 30].

The frozen catalog contains 100 unique Spotify tracks, with 25 in each region. Its retained Valence and Energy coordinates originate from a public historical dataset [18]. The repository preserves the track records but not the download date or a script reproducing their curation. The preserved catalog is a frozen stimulus set, not a fresh derivation.

6.2 Selection policy and decision trace

Both conditions exclude the current and all previously played tracks. Vibe keeps unplayed tracks in the target region and selects the smallest Euclidean distance:

$$s^* = \arg\min_{s \in Q(\bar{V}, \bar{A})} \sqrt{(V_s - \bar{V})^2 + (E_s - \bar{A})^2}.$$

A session seed resolves exact ties. If the region is exhausted, the policy searches the complete unplayed pool. Random uses the same pool and exclusions but ranks candidates by a session-seeded pseudorandom score. Catalog, playback path, exposure count, and rating questions remain constant, isolating the selection rule more closely than a comparison between different interfaces.

Because sensing is optional, a Vibe decision can use camera evidence, cardiac evidence, both, or neither. The retained summaries do not report these frequencies; the first track may also precede a completed facial reference.

Adaptation changes only the next-song ranking. The completed trajectory produces a target, region, eligible set, distance ranking, and selected track. The local record exposes the final coordinate and decision, although it cannot replay every sensing step.

In a future service, a long-term preference model could first generate a suitable candidate pool. A confidence-gated Vibe policy could then adjust the ordering for the current session while preserving an override and a visible explanation. This is a proposed deployment model, not an evaluated feature of the present system.

Every transformation has a named output: personal reference, relative coordinate, trajectory mean, catalog region, eligible set, distance, and chosen track. A service should explain a re-ranking, withhold adaptation under weak signals, and offer immediate preference-only selection.

7 User Study & Reflection

7.1 Study design and analysis

The exploratory study used a within-participant crossover with five-track Random and Vibe blocks drawn from the same frozen catalog. Two protocols reversed block order; the current interface suggests an assignment from the participant number and masks condition names. Administration can still reveal the condition, and the study was not double blind. Reversing order reduces but does not remove practice, fatigue, or carryover risks [8].

Each track plays for up to 60 seconds, with an option to rate earlier. After each track, participants answer *How much do you like this song?* and *How well did it fit your current mood?* on seven-point scales. They also select Energetic, Calm, Tense, Melancholic, or Neutral as a categorical mood report. Mood fit measures the proposed contextual match; liking distinguishes that judgment from general song preference. Listening duration, early rating, condition order, and the current implementation's selection fields are exported.

In the current protocol, the experimenter selects a masked order and pseudonymous participant number after Spotify authentication. The participant completes available references, hears and rates five tracks per condition, and downloads the local study file. Vibe uses the preceding listening window; Random ignores it. The historical software revision is unknown, so this is the present protocol rather than a reconstruction of every pilot session.

The retained data contain 15 participant/session rows under generic identifiers, each with one five-trial condition mean for mood fit and liking. Trial rows, participant characteristics, study administration, hardware, and a code revision tied to collection are unavailable. These records cannot reconstruct trial completeness, exclusions, within-session variation, or signal-path frequencies.

Paired differences are Vibe minus Random, and the participant/session five-trial mean is the unit of analysis. We report condition means and standard deviations, paired mean differences, two-sided 95% confidence intervals, standardized paired effects, and the supplied directional paired tests. Confidence intervals use the paired t distribution with 14 degrees of freedom. Figure 3 adds 20,000 fixed-seed bootstrap resamples to show the mean distribution; its interval bars remain t-based.

The archived means are rounded to one decimal place and reproduce the supplied condition summaries and paired t results to rounding. Exact signed-rank values depend on unavailable precision and tie handling, so we retain the supplied values. No dated preregistration or confirmatory threshold was available. Estimation and two-sided uncertainty are primary; the one-sided tests are reported descriptively [6, 20].

7.2 Results

Mood fit averaged 4.71 under Vibe and 4.15 under Random (Table 1). The paired difference was +0.56 rating points, with a 95% confidence interval from −0.21 to 1.33 and a standardized paired effect of $d_z = 0.40$. Eight participant/session means were higher under Vibe, one was tied, and six were lower. The retained directional paired test yielded $t(14) = 1.56$, $p = .070$; the supplied signed-rank sensitivity check yielded $W^+ = 74.0$, $p = .088$. The interval includes effects on both sides of zero.

Liking averaged 4.56 under Vibe and 4.47 under Random. The paired difference was +0.09, with a 95% confidence interval from −0.52 to 0.71 and $d_z = 0.08$. The directional tests yielded $t(14) = 0.32$, $p = .375$, and $W^+ = 58.5$, $p = .353$. No equivalence margin was specified, so the near-zero estimate does not establish equivalence.

The pilot does not answer the research question decisively. Mood-fit ratings were higher on average under Vibe Shuffle, whereas liking changed little, but neither estimate excludes zero or establishes a recommendation benefit.

Participant/session differences span both directions, with several exceeding one rating point. The retained means cannot distinguish familiarity, context, order, signal availability, or differences in the value of matching; one average should not be treated as a universal personalization effect.

7.3 Reflection, limitations, and next evaluation

The study and subsequent artifact audit clarify what the project can support. Personal references provide a local comparison point, but they do not validate the facial or cardiac mappings. A trajectory makes the decision easier to inspect, but its arithmetic mean gives more weight to periods with more coordinate changes. The matched Random condition controls catalog size, playback path, exposure count, and rating procedure; selected songs can still differ in artist, genre, language, familiarity, and popularity.

The audit replaced emotion-detection language with baseline-relative evidence, separated diagnostic coherence from selection, and defined the decision record around target, region, distance, signal source, and quality flags. These reporting changes do not strengthen the historical evidence, and no qualitative feedback archive supports participant-driven redesign.

Sixty-three deterministic tests cover catalog balance, export quality flags, facial rules, RR filtering, baseline-relative cardiac behavior, and song selection. They validate the present modules, not sensor accuracy, playback, the end-to-end study flow, or the pilot software revision.

Validity limits. Three limitations define the evidence boundary. First, the facial and cardiac mappings are fixed heuristics rather than independently validated measures. Movement may alter camera evidence, raise heart rate, and degrade RR quality at the same time, allowing one behavior to influence both axes. Short HRV windows and unmeasured respiration further constrain interpretation.

Second, the small exploratory crossover and incomplete archive prevent trial-level, order, carryover, missing-data, and subgroup analyses. The retained files contain no recruitment, ethics, hardware, or trial-level records, so those parts of the study cannot be reconstructed. One-sided tests should not be treated as confirmatory without preregistration. The first adaptive choice may precede completed references, sensing is optional, and the historical summaries do not identify the signal path behind each choice.

Third, the frozen catalog improves experimental control but does not represent a dynamic streaming service. The conditions share a catalog, not identical tracks. The system also uses event-weighted rather than time-weighted aggregation. At the exact no-signal center, current code paths disagree over whether the boundary belongs to Calm or Energetic. That edge case should be unified before another study.

Table 1: Exploratory summaries of 15 participant/session five-trial means. Confidence intervals are two-sided; retained tests use one-sided Vibe-over-Random alternatives.

Outcome	Vibe M (SD)	Random M (SD)	Difference [95% CI]	$t(14)$, p	W^+, p	d_z
Mood fit	4.71 (0.99)	4.15 (1.30)	+0.56 [−0.21, 1.33]	1.56, .070	74.0, .088	0.40
Liking	4.56 (1.18)	4.47 (0.77)	+0.09 [−0.52, 0.71]	0.32, .375	58.5, .353	0.08

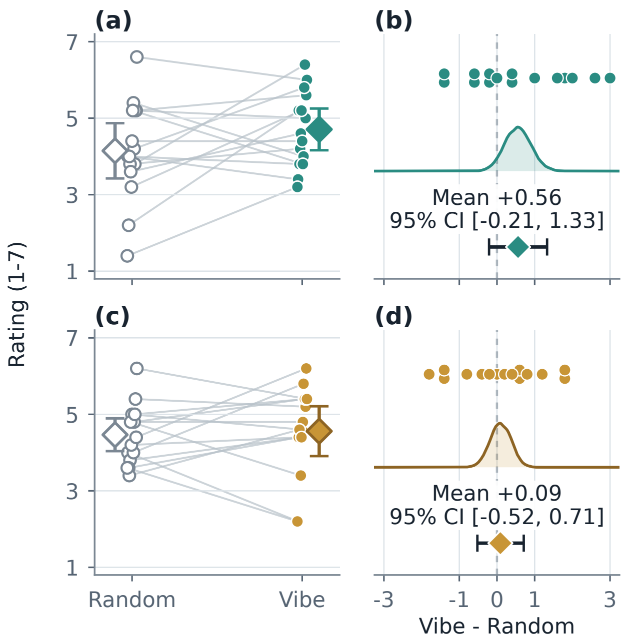

Figure 3: Paired participant/session outcomes. Panels (a)–(b) show mood fit; panels (c)–(d) show liking. Left panels connect Random and Vibe five-trial means and summarize condition means. Right panels show all paired differences, fixed-seed bootstrap mean distributions, and reported paired means with t-based 95% CIs. Both paired intervals cross zero.

Next evaluation. The next evaluation should test the frozen mappings under controlled changes in expression, lighting, posture, movement, and respiration. An instrumented pilot should require completed references, timing and missing-data flags, and all ten trial records. Camera-only, cardiac-only, fused, and Random conditions would identify the useful evidence path; time-weighted and event-weighted trajectory summaries should be compared directly.

Before a larger crossover, preregister the smallest relevant mood-fit effect, sample size, exclusions, paired and order analyses, and signal-quality rules [16]. A dynamic-catalog study should record availability, playback failures, coverage, revisions, explanations, overrides, and privacy expectations. This separates measurement failure from recommendation failure and tests whether session-aware re-ranking adds value above a preference model.

8 Conclusion & Future Work

Vibe Shuffle studies whether within-listener camera and cardiac changes can inform the next song without being presented as an emotion diagnosis. Personal references center the observations, a listening trajectory preserves their recent movement, and an explicit region-and-distance rule converts the trajectory into a reviewable catalog decision. The mechanism is specified at the level of system design and can be examined independently of the outcome claim.

The crossover does not establish improved mood fit or liking. Mood-fit estimates favored Vibe Shuffle, but both intervals included zero and the archive cannot explain the differences. The result remains an open empirical question.

If later evidence supports the policy, session-aware re-ranking could be tested for exercise, focus, relaxation, games, or accessibility settings. Real-world use would require visible uncertainty, an immediate opt-out, data-minimizing processing, and adaptation subordinate to explicit choices.

With validated mappings, complete trial records, preregistered analyses, and a dynamic candidate pool, future work can test the policy as an auditable re-ranking layer above preference-based recommendation. Vibe Shuffle leaves a clear standard for that test: moment-aware music adaptation should be measurable in its effects, inspectable in its decisions, and falsifiable in its claims.

9 Contribution Statement

Markus Baumann, Felix Hajuj, and Miriam Janner jointly contributed to the project concept, study execution, interpretation, and review of the report. Markus Baumann additionally implemented and tested the archived repository version, reproduced the participant-level statistics, and prepared the visual analysis. The available records do not preserve a finer-grained division of writing responsibilities.

Open Science and Transparency

The project repository contains source code, the frozen catalog, tests, figure scripts, participant/session means, and result-provenance records: https://github.com/eybmits/vibe-shuffle. It contains no raw trial rows, sensor streams, recruitment records, administration logs, or code revision tied to the pilot sessions. The computational tests verify the reported source tree, not the historical data-collection environment.

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